Project Report

VENDOR PERFORMANCE ANALYSIS

A data analysis project focused on vendor efficiency, profitability, and inventory management

Tools Used: Python | SQL | Power BI

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PROBLEM STATEMENT:

In this project, I worked on analysing vendor performance to understand how different vendors are affecting sales, profit, and inventory.

The main idea was to find:

- Which vendors are doing well
- Where money is getting stuck in inventory
- Which brands need promotion or pricing changes

This helps a business take better decisions related to purchasing and sales.

DATA OVERVIEW:

The data was available in multiple files like:

- Sales
- Purchases
- Vendor invoices
- Purchase prices

Since the data was scattered, I first combined everything using Python and stored it in a SQLite database. This made the analysis easier and structured.

DATA ISSUES & CLEANING:

While working on the data, I found some real-world issues:

- Some products had **zero sales** - purchased but never sold
- Some cases had **negative profit** - selling at loss
- Huge variation in **freight cost and pricing**
- Some values were extremely high (outliers)

IMPORTANT ISSUE I FACED:

While calculating unsold inventory, I got negative values.

That means: Sales > Purchase

Reason for that came to be because the sales were happening from **old stock (previous inventory)**

Instead of taking negative values I fixed it using the below steps:

- I converted negative unsold units to **0**
- Because those items are already sold

This changed the total unsold capital significantly and made the analysis more accurate.

METRICS CREATED:

To understand performance better, I created some important metrics:

- **Gross Profit** - Sales - Purchase
- **Profit Margin** - Profit / Sales
- **Stock Turnover** - Sales Quantity / Purchase Quantity
- **Unsold Inventory Value** - Unsold Units x Purchase Price

These helped in measuring both profit and inventory efficiency.

KEY FINDINGS:

Around **$9.5M was stuck in unsold inventory initially**, but after fixing calculation it became around **$2.7M**

- A few vendors contribute most of this unsold capital
- Some brands have **high profit margin but low sales** - good opportunity for promotion
- Some products are purchased but never sold - poor demand planning
- Freight cost varies a lot - possible inefficiency in logistics

These insights match what is shown in summary stats and charts in the report “Vendor Performance Report”

DASHBOARD EXPLANATION:

I created a Power BI dashboard to make insights easy to understand.

It includes:

- KPI cards (Profit, Sales, Unsold Capital)
- Top vendors and brands by sales
- Low-performing vendors
- Contribution chart (Top vendors)
- Scatter plot to identify high margin but low sales brands

This helps quickly understand where action is needed.

BUSINESS INSIGHTS:

From the analysis:

- High-margin but low-sales brands can be promoted
- Some vendors are over-supplying - causing inventory pile-up
- Business is dependent on a few vendors (risk factor)
- Bulk purchasing reduces cost but can increase inventory risk

RECOMMENDATIONS:

Based on the analysis:

- Reduce purchases from vendors with high unsold inventory
- Promote high-margin products to increase sales
- Improve demand forecasting
- Avoid overstocking
- Optimize freight/logistics cost
- Balance dependency across vendors

CONCLUSION:

This project helped me understand how real business data works.

I learned:

- How to combine multiple datasets
- How to clean and fix real-world data issues
- How to find insights that actually matter
- How to present everything in a dashboard